

INVESTOR & TECHNICAL BRIEFING

Nivix

Cross-border payments settled in seconds, not days

Solana + USDC settlement

Hyperledger Fabric compliance

GIFT City IFSC / IFSCA

Company: NivixPe Private Limited

Incubation: Bennett University Hatchery

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1. Executive Summary

Nivix is a cross-border payments platform that settles international transfers in **seconds instead of days**, at a cost of **under \$1** instead of the \$25–45 typically charged by bank wires. It does this by using **USDC** — a fully regulated, dollar-backed stablecoin issued by Circle — as the settlement layer between two licensed local payout partners, one in the sender's country and one in the recipient's.

Two things make this legally and technically sound, and both are deliberate design choices, not accidents:

- **Nivix never issues its own token.** Every dollar of value on-chain is real, audited USDC. This avoids the unlicensed-stablecoin problem that sinks most "crypto remittance" projects.
- **Nivix operates under GIFT City IFSC**, a jurisdiction India's own regulator (IFSCA) treats as a non-resident zone for foreign-exchange law. This is what makes a stablecoin-settled corridor legal for us when it would not be legal from mainland India directly.

The one-line pitch: Licensed fiat partners collect and pay out local currency on each end. We hold our own pre-funded USDC treasury in the middle and move it on Solana in about two seconds. We own the treasury, the routing logic, and the compliance ledger — the pieces that create defensibility and margin.

This document has two audiences in mind. If you are evaluating the business, Sections 1–3 and 6–10 are the ones that matter. If you or your technical advisor want to verify the engineering underneath the pitch, Sections 4, 5, and Appendix A go deep enough to do that. Section 5 in particular corrects a simplification from our earlier investor memo — we'd rather you hear the precise version from us than discover the gap yourselves.

2. The Problem

Cross-border retail and SME payments still run on infrastructure designed in the 1970s (SWIFT correspondent banking):

Metric	Traditional banking / SWIFT	Modern fintech (Wise, Remitly)
Settlement time	3–5 business days	1–2 days (still batches through banking hours)
Fee on a \$100 transfer	\$25–45	\$5–15
Transparency	Opaque — no visibility into where funds sit mid-transfer	Partial tracking, still opaque internally
Availability	Banking hours only	24/7 front-end, but still bank-hour settlement underneath

The bottleneck is always the same: money has to physically move between correspondent banks, each taking a cut and a day. Stablecoins solve the movement problem — a USDC transfer settles on Solana in about two seconds for a fraction of a cent — but only if the fiat legs on either end are handled by properly licensed entities. That licensing question, not the technology, is why most "crypto for remittance" projects stall. It's also why we built the legal structure (Section 6) before the payment rails.

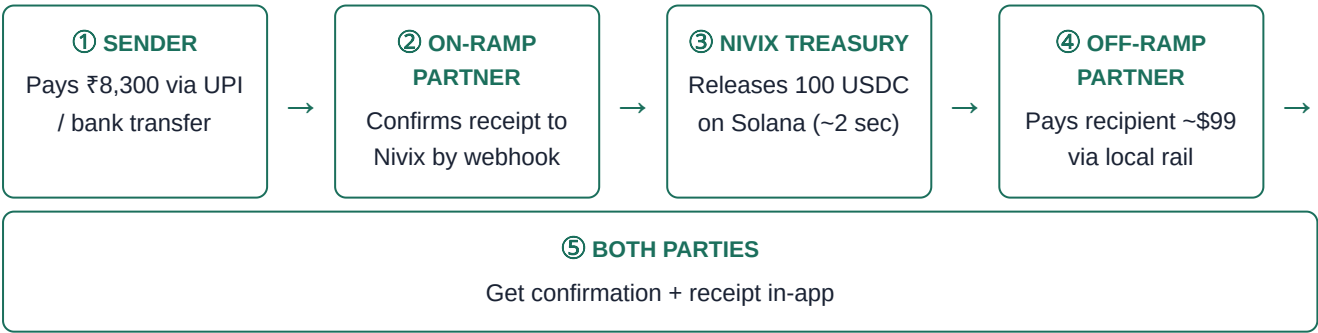
3. How Nivix Moves Money

3.1 Who does what

Entity	Holds	Role
On-ramp fiat partner (e.g. Razorpay/Cashfree in India)	Local fiat only — INR, never crypto	Collects the sender's payment, confirms it to Nivix via webhook
Nivix treasury	Pre-funded USDC pool on Solana	Releases USDC the instant fiat is confirmed; receives USDC back on reverse flows; owns the compliance ledger
Off-ramp fiat partner (destination country)	Local fiat only — USD/EUR/etc., never crypto	Pays the recipient's bank account in local currency (see Section 5 for exactly how this leg works)
Hyperledger Fabric (multi-organisation)	KYC and compliance records	Immutable, private audit trail — no single party, including Nivix, can alter a record alone
Solana (public chain)	USDC settlement transactions	Fast, cheap, publicly verifiable money movement and proof of reserves

Design principle: fiat partners never touch crypto, and Nivix never touches customer fiat directly beyond the confirmation webhook. This separation of duties is what keeps each hop within existing licensing regimes instead of requiring Nivix itself to be licensed as a money transmitter in every country.

3.2 Step by step — India → USA example



Step	What happens	Who	Time
①	Sender pays local currency via UPI/bank transfer	Licensed, locally-regulated on-ramp partner	Instant
②	Payment confirmed to Nivix, signature verified	Nivix API	Seconds

③	Nivix releases USDC from its own treasury on Solana	Nivix treasury wallet	~2 seconds
④	Off-ramp partner pays the recipient in local currency	Licensed off-ramp partner (see Section 5)	Minutes–hours, rail-dependent
⑤	Both sides see status + receipt	Nivix app	Instant

3.3 Why a treasury instead of a pass-through

An alternative design would have Nivix simply forward whatever the on-ramp partner sends, without holding its own balance. We rejected that:

No treasury (pass-through only)	Own treasury (our choice)
Settlement waits on the partner's own conversion speed	USDC is already in our wallet — release is instant
We don't control the spread	We control the full margin on both legs
A partner outage stalls the whole transfer	Treasury float gives us buffer and independence

3.4 Keeping the treasury funded

The treasury is a float, not a per-transaction top-up:

TWO-WAY CORRIDOR (India $\neq$ USA):

India→USA leg: ₹ in (treasury's INR side grows) | USDC out (treasury shrinks)
 USA→India leg: \$ in via reverse flow (USDC side grows) | ₹ out (INR side shrinks)
 NET EFFECT: balanced two-way traffic self-balances the treasury automatically.

ONE-SIDED CORRIDOR (e.g. 90% India→USA, 10% reverse):

Accumulated INR is periodically converted to USDC (weekly/monthly, not per-transfer)
 Auto-trigger: buy USDC when treasury capacity < 30%. Alert ops if < 10%.

4. Technical Architecture

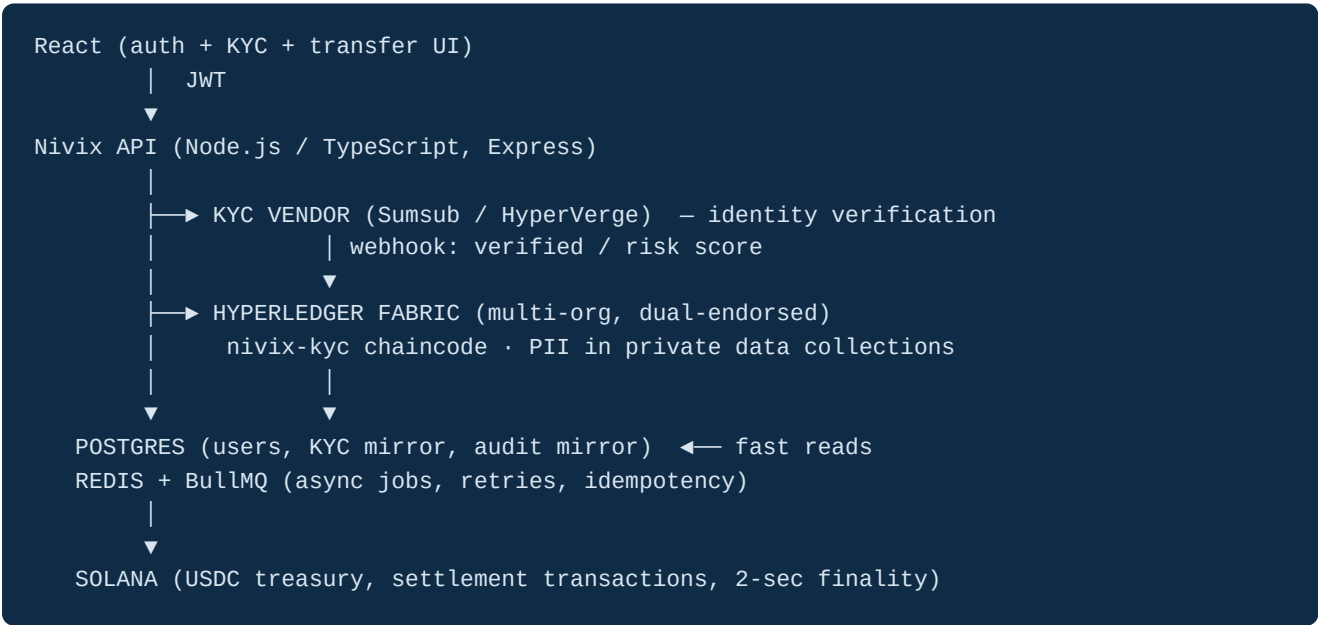
This section is for readers (or your technical advisors) who want to verify the engineering, not just the pitch.

4.1 Why two blockchains, not one

Concern	Chain used	Why not the other one
Money settlement (must be publicly verifiable)	Solana — USDC transfers, treasury balance, proof of reserves	A private ledger can't prove solvency to the outside world; a public one can
Compliance records (KYC, PII, risk decisions)	Hyperledger Fabric , multi-organisation	Personal identity data cannot legally sit on a public chain anyone can read

The fraud-safety property that matters to a diligence team: Fabric is deployed multi-org (Nivix + an independent endorsing organisation). Every KYC write requires endorsement from *both* organisations before it commits. Nivix alone cannot alter or fabricate a compliance record — the same class of control failure that sank FTX is structurally closed off here, because no single party controls the ledger.

4.2 System diagram



4.3 Repository & stack

Monorepo layout

apps/api — Express + TypeScript + Prisma; auth, KYC endpoints, webhook intake

Core technologies

Settlement: Solana + Circle USDC

apps/worker — BullMQ job processor; owns the Fabric gateway client

apps/web — React + Material UI frontend

packages/shared — Zod schemas and shared types used by both API and worker

Compliance ledger: Hyperledger Fabric (multi-org), Go chaincode

Backend: Node.js, TypeScript, Prisma ORM

Data: PostgreSQL (operational + fast-read mirror), Redis (jobs/queues)

Auth: JWT, role-based access (user / compliance / admin)

4.4 Build-for-the-swap: local now, cloud later

The current AWS account is mid-verification, so Phase 1 is built entirely on local Docker infrastructure — but every hosting dependency sits behind a small interface, so migrating to managed cloud services later is a configuration change, not a rewrite:

Concern	Local (now)	AWS (when verified)
Compliance ledger	Fabric test-network (Docker, Org1+Org2)	AWS Managed Blockchain (Nivix + genuinely independent endorser)
Database	Postgres container	RDS Postgres
Cache / queues	Redis container	ElastiCache
Secrets	Gitignored .env behind a SecretsProvider interface	AWS Secrets Manager
Encryption keys	Local master key behind an Encryptor interface	AWS KMS

Caveat we hold ourselves to: running Org1+Org2 both on one machine proves the dual-endorsement **mechanics** — it is not yet tamper-proof against Nivix itself. That property arrives when the independent endorser is onboarded onto separate infrastructure at the AWS-migration step, which is already in progress as a parallel workstream.

5. Settlement Reality: What Circle Actually Does

This section exists because an earlier internal explanation of the off-ramp leg was oversimplified. We're correcting it here deliberately, because it's exactly the kind of gap a technical due-diligence reviewer would probe for — better it comes from us.

The simplification we're correcting: Our July memo described "Circle Mint" as the off-ramp partner that pays the recipient directly. Having read Circle's own API documentation closely, that is not what Circle Mint does.

5.1 What Circle Mint actually is

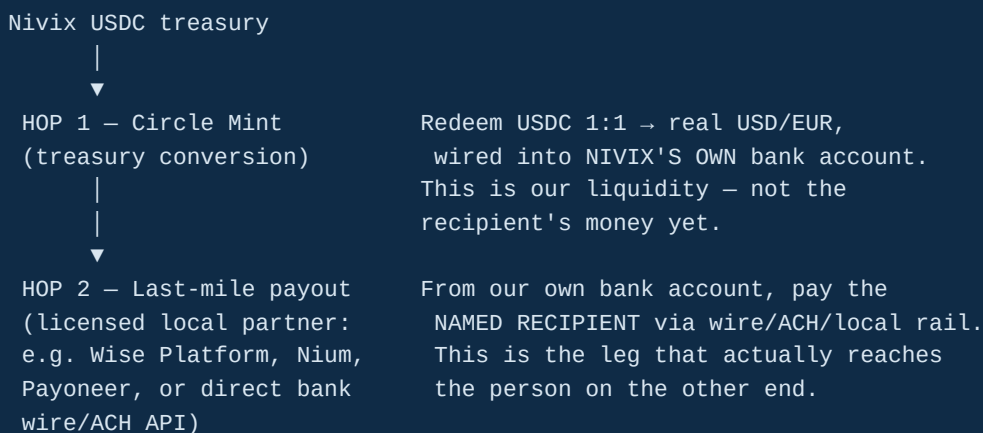
Circle Mint lets an approved business redeem its own USDC 1:1 for real US dollars (or EURC for euros), wired into **a bank account that business has already pre-registered with Circle**. Circle's own quickstart guide shows the redemption call requires a destination object referencing an account ID created in an earlier step — there is no field for an arbitrary third party's bank details:

```
"destination": {
  "type": "wire",
  "id": "9d1fa351-b24d-442a-8aa5-e717db1ed636"  ← must be OUR pre-linked account
}
```

In other words: **Circle Mint turns our own USDC into our own dollars**. It is treasury infrastructure for Nivix, not a payout API to Nivix's customers. Access also requires Circle's own KYB/sanctions review (days to weeks) and is limited to wire/SEPA rails at participating banks in eligible regions.

5.2 The corrected model: two hops, not one

The international leg of a transfer therefore needs **two separate, licensed steps**, not one:



Why this is actually a strength, not a weakness: no single partner ever touches both the crypto leg and the fiat-to-recipient leg. That separation is a compliance feature — it mirrors the same multi-party

control principle we apply on the Fabric side (Section 4.1), just applied to the money rails instead of the ledger.

5.3 What's already proven on the India side

On the India (INR) side, this two-hop concern doesn't arise the same way: our current integration with **Cashfree Payouts** bundles both jobs into a single API — it holds settlement funds and pays named beneficiaries directly (bank transfer, or a claim-link "Cashgram" the recipient redeems to a bank account or UPI of their choice). This is the flow already demonstrated end-to-end in our proof-of-concept: sender pays INR → on-chain settlement → recipient receives INR in their own account.

What remains open is the **equivalent last-mile partner for each destination country** — that integration work is scoped as Phase 2 (Milestone M2/E5) of the build plan and has not started yet. It is a scope item, not a design flaw: the treasury, the Solana settlement rail, and the compliance ledger are all corridor-agnostic and don't change when we add a country.

5.4 How to say this to a technical reviewer

"Circle Mint is our treasury conversion layer — it turns our own USDC into real, regulated dollars in our own account, 1:1. The last mile — actually paying the named recipient — is a separate, licensed payout integration per corridor: Cashfree today in India, and an equivalent partner (Wise Platform, Nium, or a direct bank wire/ACH API) for each new country we add. Two regulated hops, each auditable independently."

6. Legal & Regulatory Structure

Aspect	Status
Legal home	GIFT City IFSC, Gujarat — regulated by IFSCA
FEMA status	IFSC is treated as a non-resident jurisdiction — mainland India's foreign-exchange restrictions on crypto don't apply inside it
Entry route	IFSCA FinTech Regulatory Sandbox — explicitly permits a stablecoin-settled pilot with real, limited customers
Report submitted to IFSCA	Yes
Company / incubation	NivixPe Private Limited, Bennett University Hatchery
Tax treatment	Up to a 20-year tax holiday available in GIFT City under Budget 2026 provisions
KYC / AML	KYC vendor (Sumsub/HyperVerge) for verification; Hyperledger Fabric (multi-org) for the immutable compliance record
Data protection	DPDPA 2023-aligned; PII stored only in Fabric private data collections, never on the public chain

The rule that keeps the whole model legal: Nivix orchestrates and records. Licensed fiat partners handle local currency on each end. Nivix itself holds USDC, never customer fiat. The IFSCA sandbox is what permits that stablecoin settlement inside its jurisdiction.

This is a regulatory pathway, not a loophole for mainland retail INR flows — money originating or terminating outside the IFSC still needs the standard LRS/AD-bank route or a FIU-registered VDASP partner for its domestic leg.

7. Revenue Model

Stream	Source	Typical margin
Platform fee	Flat % per transaction	0.3–0.5%
FX spread	Difference between the locked user rate and the market rate	0.2–0.5%
Treasury yield	Idle USDC in the treasury earns interest (Circle offers ~4.5% APY on qualifying balances)	~4.5% APY on idle float
Volume incentives	Fiat partner rebates at volume tiers	Variable

Blended take rate: 0.5–1.0% per transaction, versus 5–10% typical for traditional remittance.

Illustrative economics at \$1M monthly volume (10,000 transactions @ \$100 avg):

Gross margin per transaction:	\$0.60 – \$1.00
Monthly revenue:	\$6,000 – \$10,000
Annualised:	\$72,000 – \$120,000

(Illustrative only – does not include treasury yield or partner rebates.)

8. Build Status & Roadmap

8.1 What's demonstrated today

- End-to-end proof-of-concept: sender pays INR → on-chain token settlement → recipient receives INR via Cashfree payout
- KYC flow (submission, verification, admin review) exercised in the legacy prototype
- Hyperledger Fabric running 2-organisation locally, KYC chaincode deployed
- Frontend transfer wizard: KYC check → recipient details → amount → processing → receipt
- Legal report submitted to IFSCA

8.2 What's actively being rebuilt (Phase 1 — identity & compliance backbone)

The prototype proved the mechanics; we are now rebuilding the identity and compliance layer properly, in a clean TypeScript monorepo, before wiring money movement back in:

- **DONE** Clean monorepo, CI, local Docker infrastructure (Postgres + Redis)
- **DONE** Auth (register / login / JWT / role-based access), tested end-to-end
- **DONE** KYC vendor integration path, PII envelope encryption, audit logging
- **DONE** Hyperledger Fabric write path via the official Fabric Gateway SDK — KYC approval is recorded on-chain and mirrored to Postgres for fast reads, replacing the prototype's silent-failure write path
- **NEXT** USDC settlement on Solana (Milestone M2)

8.3 Roadmap

Phase	Delivers
Phase 1 — Identity & compliance (in progress)	Auth, KYC vendor integration, multi-org Hyperledger Fabric, audit trail
Phase 2 — Money movement	USDC settlement on Solana, on-ramp integration, off-ramp per-corridor (the two-hop model in Section 5), quotes/FX
Phase 3 — Sandbox pilot	Deploy into the IFSCA sandbox, limited live transactions, regulator reporting
Phase 4 — Scale	Additional corridors, additional partners, progression toward full IFSCA licensing

9. Competitive Landscape

	Traditional (SWIFT / banks)	Wise / Remitly	Nivix
Settlement time	3–5 days	1–2 days	~2 seconds (on-chain leg)
Fee	\$25–45	\$5–15	< \$1 (on-chain leg)
Transparency	Opaque	Partial tracking	Publicly verifiable on-chain settlement + auditable compliance ledger
Availability	Business hours	24/7 front-end	24/7, gated only by fiat-partner rail hours on each end

Note: the sub-\$1, ~2-second figures describe the on-chain settlement hop specifically. End-to-end time still includes the fiat collection and payout legs on either side (Section 3.2), which run at the speed of the local rail used.

10. Risk Management

Risk	Mitigation
Treasury depletion on one-sided corridors	Auto-rebalance trigger at 30% capacity; ops alert at 10%
USDC de-peg	Circle is the most regulated stablecoin issuer, backed by US Treasuries, audited monthly; real-time monitoring; ability to pause instantly
Payout partner outage	Provider interfaces are swappable per corridor; multiple partners qualifiable per corridor
Regulatory change	IFSCA sandbox status gives legal cover; proactive compliance posture; corridor-specific licensing added as needed
Fraud / money laundering	Mandatory KYC, sanctions screening, transaction limits, multi-org Fabric ledger that no single party can alter
Key compromise	Treasury secured with multisig + HSM in production; secrets held in a vault, never in source control; least-privilege access

11. Investor FAQ

Where does the money actually come from?

From our pre-funded USDC treasury. When a sender pays local currency, that currency lands in our bank account via the on-ramp partner. We release the equivalent USDC from our own Solana treasury. The treasury refills from two-way traffic plus periodic conversion of accumulated fiat — see Section 3.4.

What does Nivix actually own and control?

The USDC treasury, the orchestration and routing layer, and the compliance engine (the Hyperledger audit trail). We do not hold customer fiat directly beyond a payment's confirmation webhook, we do not issue our own token, and we do not carry unhedged FX risk.

How do you make money?

Platform fee (0.3–0.5%) plus FX spread (0.2–0.5%) plus treasury yield on idle USDC. See Section 7 for the full model and illustrative economics.

Why Solana instead of just using SWIFT?

Speed (seconds versus days), cost (fractions of a cent versus \$25–45), and a publicly verifiable settlement trail instead of opaque internal bank records — and it runs 24/7, not on banking hours.

Why Hyperledger as well as Solana?

They do different jobs. Solana is public — it's how anyone can verify our treasury reserves and settlement history. Hyperledger is private — it's where KYC and personal data live, visible only to authorized parties. Public money movement plus private compliance is the combination that lets us be both transparent and legally compliant with data protection law.

Does Circle handle the whole off-ramp?

No — see Section 5. Circle Mint converts our own USDC into our own fiat. Actually paying the named recipient is a second, separate licensed integration per corridor (Cashfree today in India; an equivalent partner needed for each new country).

Do you need to refill the treasury before every transaction?

No. Two-way corridors self-balance automatically. One-sided corridors are rebalanced periodically (weekly/monthly), with automatic threshold alerts — never per-transaction.

What if USDC loses its peg?

USDC is the most heavily regulated major stablecoin — backed 1:1 by US Treasuries and cash, audited monthly. A de-peg event would be an industry-wide shock, not a Nivix-specific one. We monitor continuously and can pause settlement instantly if needed.

What is the moat here?

First-mover regulatory positioning in GIFT City for stablecoin-settled remittance; a dual-blockchain compliance design that is hard to replicate quickly and that regulators respond well to; a treasury model that gives genuinely instant settlement instead of waiting on a partner's own conversion speed; and a routing engine designed from day one to add corridors without re-architecting.

Appendix A — Technology Stack

Layer	Technology	Purpose
Settlement	Solana + Circle USDC	Fast, low-cost, publicly verifiable money movement
Compliance ledger	Hyperledger Fabric (multi-org)	Private, tamper-resistant KYC and audit records
Backend	Node.js, TypeScript, Express, Prisma, BullMQ	API surface and asynchronous job processing
Database	PostgreSQL	Operational data and a fast-read mirror of Fabric state
Queue / cache	Redis + BullMQ	Job queue for Fabric writes, webhooks, retries
Frontend	React + Material UI	Customer-facing application
On-ramp (India)	Razorpay / Cashfree	INR collection
Off-ramp (India)	Cashfree Payouts	INR payout to named recipients (bank transfer or claim-link)
Treasury conversion	Circle Mint	1:1 USDC/EURC redemption into Nivix's own bank account (Section 5)
KYC verification	Sumsub / HyperVerge	Identity verification vendor
Hosting	Local Docker today; AWS (Managed Blockchain, RDS, ElastiCache) planned	Infrastructure, migrated config-only per Section 4.4

Appendix B — Glossary

USDC

A stablecoin issued by Circle, redeemable 1:1 for US dollars and backed by cash and short-term US Treasuries.

Solana

A public blockchain used here purely as a fast, cheap settlement rail for USDC — not for anything requiring privacy.

Hyperledger Fabric

A private, permissioned blockchain framework used for the compliance ledger. "Multi-org" means more than one independent organisation must approve (endorse) each write.

On-ramp / off-ramp

The fiat entry and exit points of a transfer — the licensed local partners who collect and pay out local currency.

IFSC / IFSCA

India's International Financial Services Centre (GIFT City, Gujarat) and its regulator, the International Financial Services Centres Authority — a jurisdiction treated as non-resident for Indian foreign-exchange law.

Circle Mint

Circle's business product for redeeming USDC/EURC 1:1 into a pre-registered bank account belonging to the redeeming business (see Section 5).

KYC / KYB

Know Your Customer / Know Your Business — identity and entity verification required before onboarding.

Endorsement (Fabric)

The requirement that a transaction be approved by specified organisations' peers before it is committed to the ledger.